

Cloud Computing: Serverless Architecture

A Comprehensive Study Guide for University Students

1. What is Serverless Computing?

Serverless computing is a cloud execution model where the provider dynamically manages server allocation and provisioning. Developers write and deploy code without managing underlying infrastructure. The term "serverless" does not mean there are no servers - it means the developer does not need to manage them. The provider handles scaling, patching, and capacity planning automatically. Key services include AWS Lambda for compute, API Gateway for HTTP endpoints, and DynamoDB for database operations.

2. Function as a Service (FaaS)

AWS Lambda is the most widely adopted FaaS platform. It executes code in response to events such as HTTP requests via API Gateway, file uploads to S3, or messages in SQS queues. Lambda supports Node.js, Python, Java, Go, and .NET runtimes. Functions scale horizontally and automatically - each invocation runs in an isolated execution environment. Key benefits include pay-per-use pricing with no idle costs, automatic scaling from zero to thousands of concurrent executions, and built-in fault tolerance with automatic retries.

3. Event-Driven Architecture

Modern cloud applications rely on event-driven patterns where services communicate through events rather than direct synchronous calls. Amazon SQS (Simple Queue Service) provides fully managed message queues that decouple producers from consumers, ensuring reliable delivery even under load spikes. Amazon SNS (Simple Notification Service) distributes messages to multiple subscribers via publish-subscribe patterns. Amazon EventBridge routes events between AWS services, SaaS applications, and custom code based on configurable rules.

4. API Gateway

Amazon API Gateway creates, publishes, and secures APIs at any scale. It acts as the front door for serverless applications, handling authentication via Amazon Cognito, request validation, rate limiting, response transformation, and caching. API Gateway integrates natively with AWS Lambda, enabling RESTful and HTTP APIs that scale with usage. JWT authorizers validate tokens from Cognito User Pools for protected endpoints.

5. Infrastructure as Code (IaC)

Infrastructure as Code defines cloud resources using declarative configuration files rather than manual console actions. AWS CDK (Cloud Development Kit) allows developers to define infrastructure using TypeScript, Python, or Java. CDK synthesizes CloudFormation templates for deployment, providing version control, repeatable deployments, and code review for infrastructure changes. This reduces human error and enables collaboration.

6. DynamoDB: Serverless NoSQL Database

Amazon DynamoDB is a fully managed NoSQL database with single-digit millisecond latency at any scale. Data is organized into tables with partition keys for horizontal scaling. It supports on-demand and provisioned capacity modes. DynamoDB Streams enable event-driven architectures where database changes trigger Lambda functions. Features include global tables for multi-region replication, point-in-time recovery, and encryption at rest.

7. Serverless Best Practices

Keep functions small and single-purpose. Use environment variables for configuration. Store secrets in AWS Secrets Manager. Implement error handling with try/catch and Dead Letter Queues (DLQ). Monitor all functions with CloudWatch Logs, metrics, and alarms. Set appropriate timeouts and memory limits to control costs. Use reserved concurrency to protect downstream resources. Implement idempotency for retried operations. Follow the principle of least privilege with IAM roles.

8. Security Considerations

Authentication via Amazon Cognito provides user management and JWT-based authorization. IAM roles grant least-privilege access between services. S3 buckets use private access with presigned URLs for uploads. API keys are stored in AWS Secrets Manager, never in source code. All data is encrypted at rest and in transit using TLS. CloudTrail provides audit logging for all API calls across the AWS account.

*Study material generated for AI Study Planner demonstration.
All concepts align with AWS Well-Architected Framework principles.*